

Explaining and Predicting Oil Prices Using OPEC Communication



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Research Objective

- Explain in-sample oil price movements and assess out-of-sample predictability using a combination of:
 - Textual data from OPEC communications
 - Numerical oil market fundamentals

Basic Idea

- OPEC communication conveys information about supply intentions, expectations, and market conditions
- We examine whether textual signals from OPEC add explanatory and predictive power beyond standard numerical variables

Textual Corpus

- OPEC official speeches and press releases
- Time span: 2003-2025
- 391 speeches and 572 press releases

Numerical Data

- Oil supply and demand
- Oil inventories and consumption
- Weather Data
- Etc.

Approach

- Extract textual features from OPEC communication
- Combine text-based measures with numerical oil market fundamentals

Models

- Penalized regression models like Lasso & Ridge
- Hybrid models combining text with time series:
 - ARIMA-type specifications (model oil price dynamics)
 - GARCH-type models (capture oil price volatility)
- (Additional ML approaches)

Closest Reference Papers

- Känzig (2020): Shows that OPEC announcements generate oil supply news shocks with significant price effects.
- Nazer & Pescatori (2022): Finds that OPEC announcements affect oil price volatility and studies OPEC communication using text analysis.
- Quint & Venditti (2020): Evaluates the effectiveness of OPEC+ supply cuts on oil prices via structural counterfactual analysis.